

BACKLOG BUDDY ? STUDY MATERIAL

Design and Analysis of Algorithms ? Unit-wise Lecture Notes

Subject Code: CS403 Semester: 4

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Condensed notes covering all five units: Asymptotic Analysis, Divide and Conquer, Greedy Algorithms, Dynamic Programming, Graph Algorithms and NP-Completeness.

Unit 1: Asymptotic Analysis

- Definition, key concepts and significance of Asymptotic Analysis
- Important diagrams / derivations / algorithms for Asymptotic Analysis
- Solved examples and previous-year questions on Asymptotic Analysis

Unit 2: Divide and Conquer

- Definition, key concepts and significance of Divide and Conquer
- Important diagrams / derivations / algorithms for Divide and Conquer
- Solved examples and previous-year questions on Divide and Conquer

Unit 3: Greedy Algorithms

- Definition, key concepts and significance of Greedy Algorithms
- Important diagrams / derivations / algorithms for Greedy Algorithms
- Solved examples and previous-year questions on Greedy Algorithms

Unit 4: Dynamic Programming

- Definition, key concepts and significance of Dynamic Programming
- Important diagrams / derivations / algorithms for Dynamic Programming
- Solved examples and previous-year questions on Dynamic Programming

Unit 5: Graph Algorithms and NP-Completeness

- Definition, key concepts and significance of Graph Algorithms and NP-Completeness
- Important diagrams / derivations / algorithms for Graph Algorithms and NP-

Completeness

- Solved examples and previous-year questions on Graph Algorithms and NP-Completeness

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.